fast_style_transfer

March 10, 2026

[9]:
```
''''''from google.colab import drive
drive.mount("/content/drive")
'''
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call
drive.mount("/content/drive", force_remount=True).
```

[10]:
```
import numpy as np
import pandas as pd
import tensorflow as tf
import matplotlib.pyplot as plt
import tensorflow_hub as hub
import time
import os
from PIL import Image
import shutil
```

[11]:
```
#/content/drive/My Drive/gan/gan-getting-started
#base_path = "/content/drive/My Drive/gan/gan-getting-started"
base_path = "/gan-getting-started"
content_path = os.path.join(base_path, "photo_jpg")
style_path = os.path.join(base_path, "monet_jpg")
output_path = "/images1"
image_size = 256
num_outputs = 7000
```

[12]:
```
# show #count random photos
def sample(photos, count):
    photos = photos[np.random.choice(len(photos), count)]
    for photo in photos:
        img = tf.keras.preprocessing.image.load_img(photo)
        plt.imshow(img)
        plt.show()
```

[13]:
```
style_images = np.array([os.path.join(style_path, path) for path in os.
 ↪listdir(style_path)])
np.random.shuffle(style_images)
print(style_images[:10])
```

```
['/content/drive/My Drive/gan/gan-getting-started/monet_jpg/7341d96c1d.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/e9f5563817.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/c67ba2060c.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/676a5a4c2e.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/89d970411d.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/2cca56415e.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/fb93438ff9.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/c68c52e8fc.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/cd6623d07d.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/monet_jpg/8314acfd35.jpg']
```

[14]:
```
content_images = np.array([os.path.join(content_path, path) for path in os.
 ↪listdir(content_path)])
np.random.shuffle(content_images)
print(content_images[:10])
```

```
['/content/drive/My Drive/gan/gan-getting-started/photo_jpg/710afec54e.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/92a77e056a.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/b8d30c2ede.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/a542e05538.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/e91a6a79ca.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/43ef5533dd.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/81a5890567.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/d3b6567912.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/90e1c5d57a.jpg'
 '/content/drive/My Drive/gan/gan-getting-started/photo_jpg/f54dd83443.jpg']
```

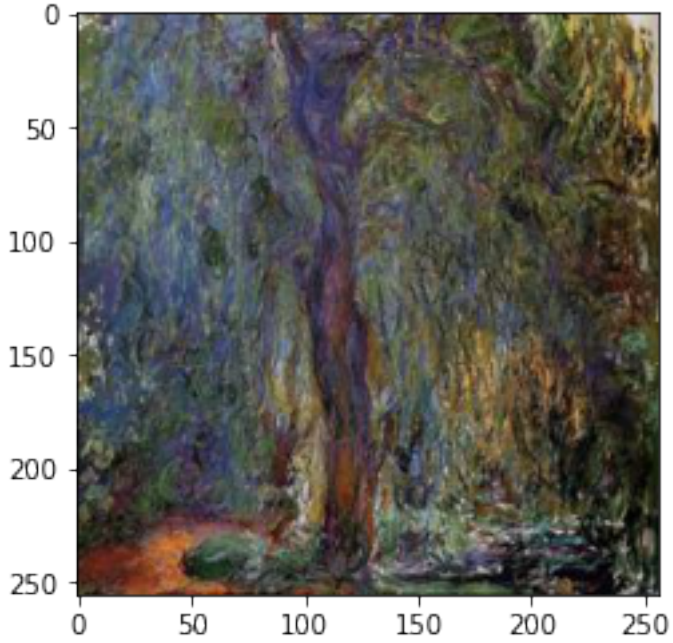
[17]:
```
len(style_images)
```

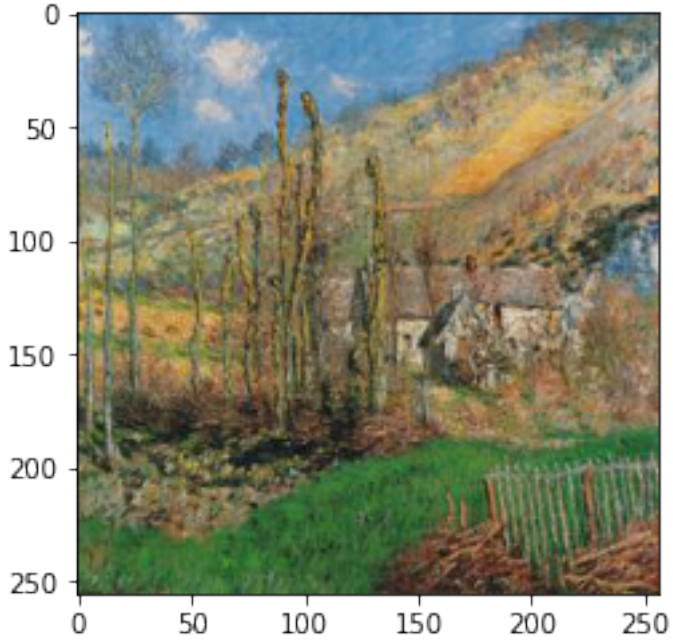
[17]: 300

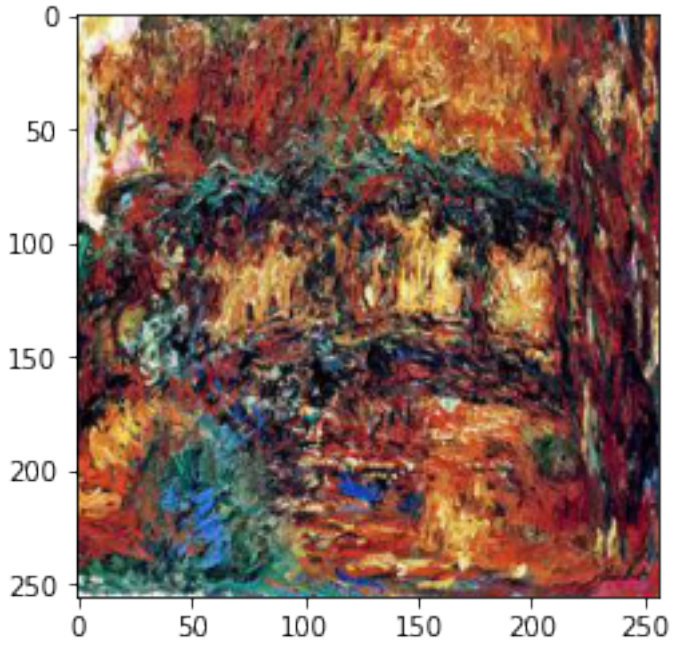
[18]:
```
len(content_images)
```

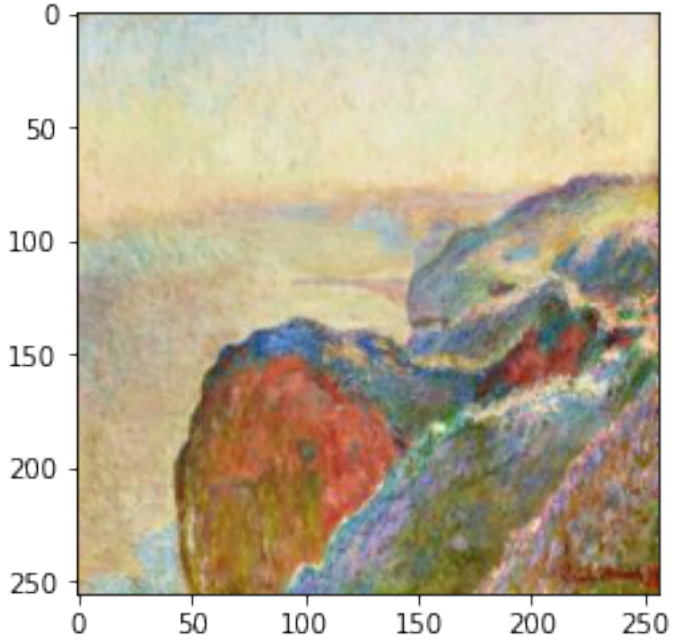
[18]: 7038

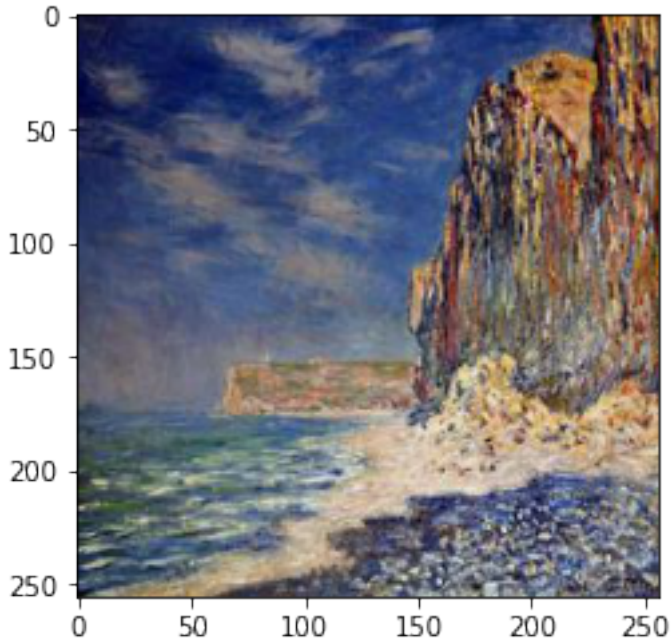
[19]:
```
sample(style_images, 5)
```


0
50
100
150
200
250
0
50
100
150
200
250

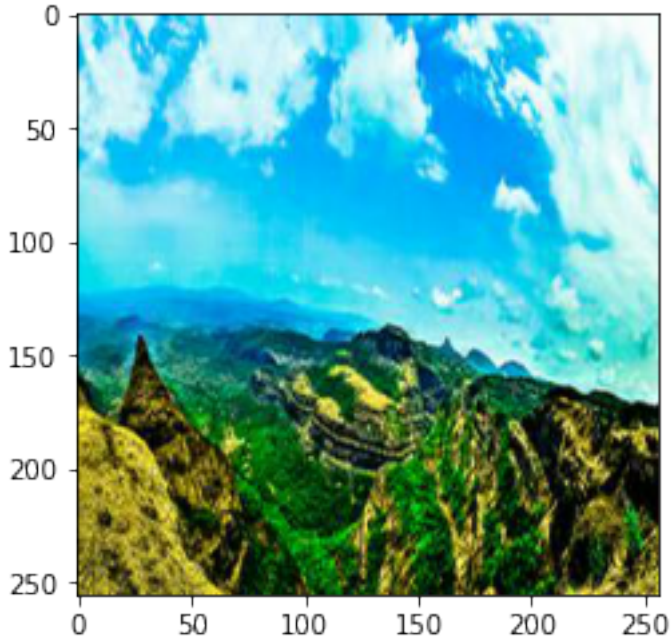

0
50
100
150
200
250
0
50
100
150
200
250

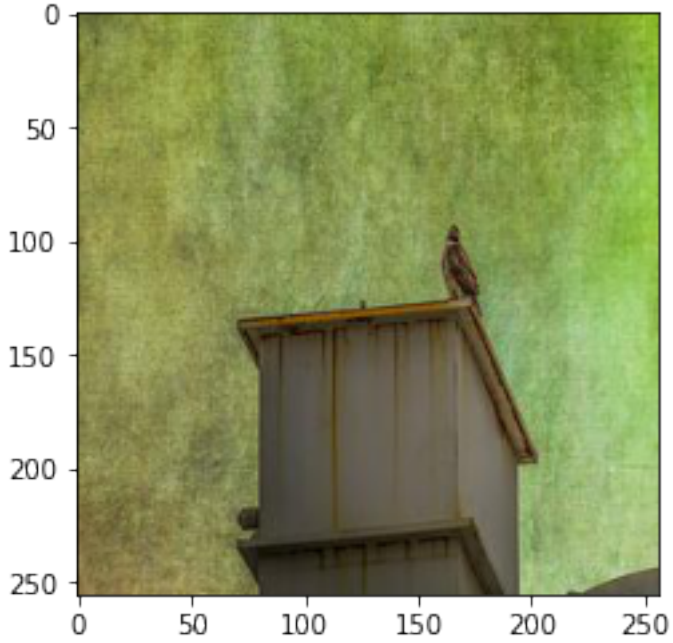

0
50
100
150
200
250
0
50
100
150
200
250

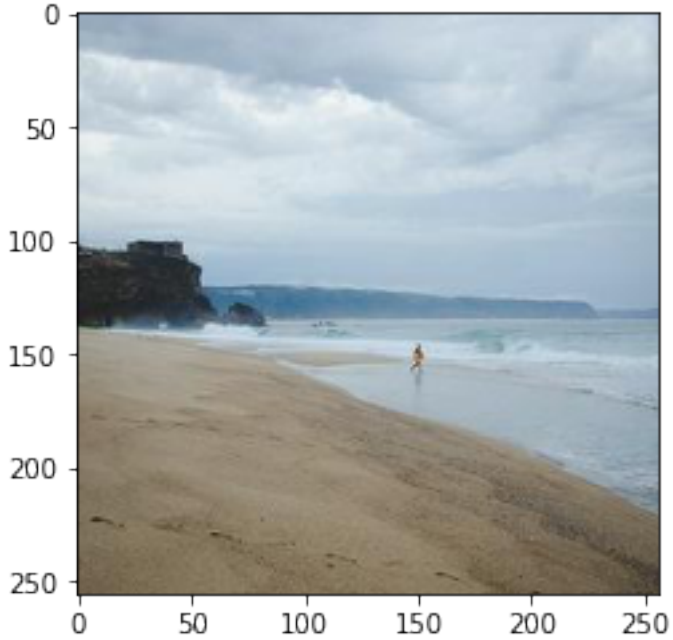

0
50
100
150
200
250
0
50
100
150
200
250

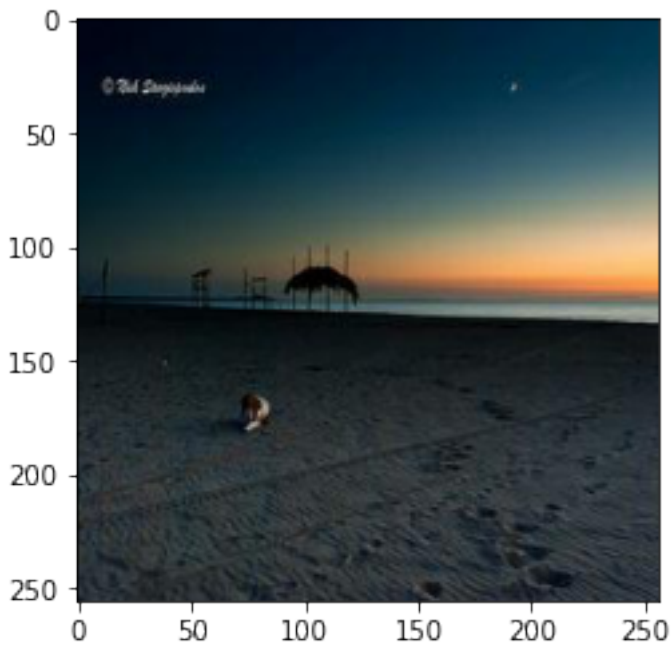


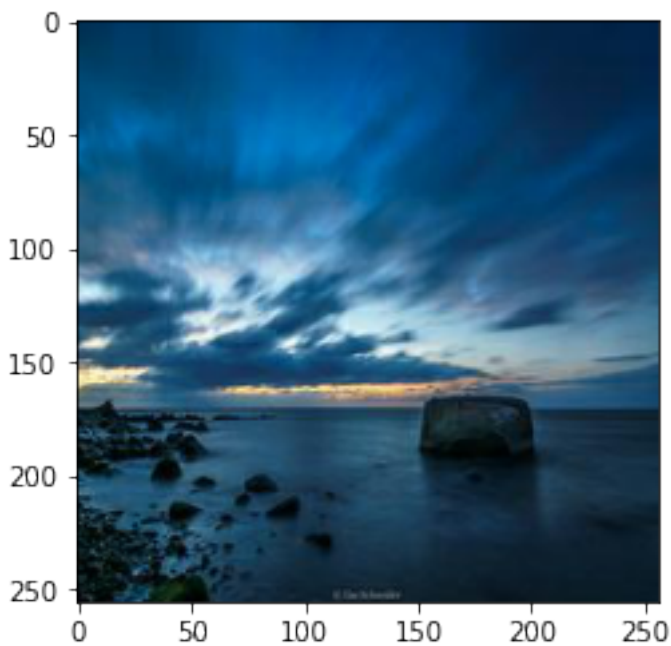
[20]:
```
sample(content_images, 5)
```




0
50
100
150
200
250
0
50
100
150
200
250


0
50
100
150
200
250
0
50
100
150
200
250





Load pretrained Model

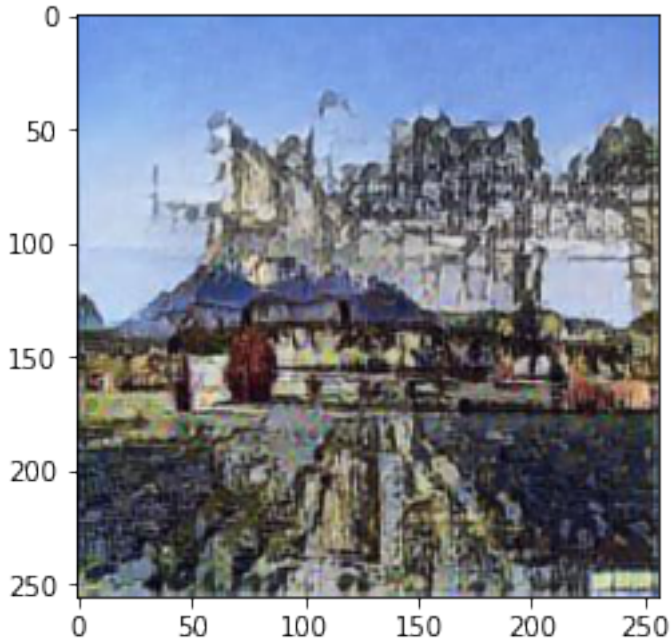
[21]:
```
hub_module = hub.load('https://tfhub.dev/google/magenta/
↪arbitrary-image-stylization-v1-256/2')
```

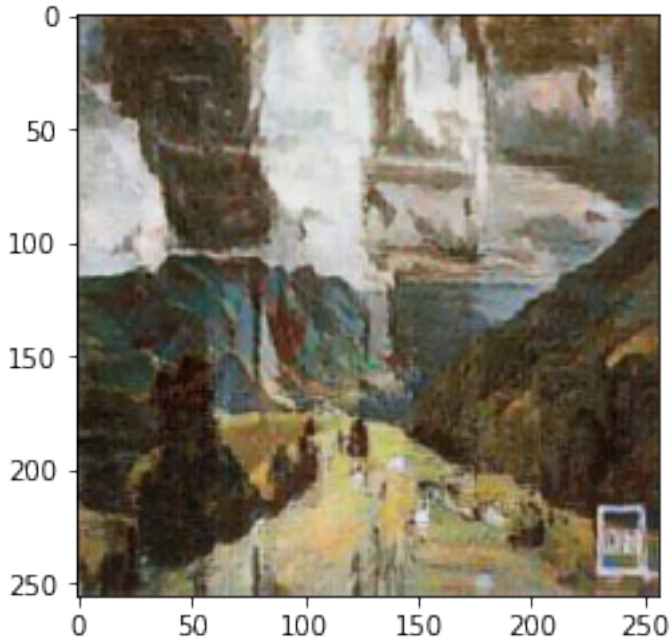
[23]:
```
#generate output
begin = time.time()
for i in range(num_outputs):
    content_image_path = content_images[np.random.randint(len(content_images))]
    style_image_path = style_images[np.random.randint(len(style_images))]
    content_image = plt.imread(content_image_path)
    style_image = plt.imread(style_image_path)
    content_image = content_image.astype(np.float32)[np.newaxis, ...] / 255.
 ↪#(2,) --> (1,3) increase dimension [1,2,3] --> [[1,2,3]]
    style_image = style_image.astype(np.float32)[np.newaxis, ...] / 255.
    style_image = tf.image.resize(style_image, (image_size, image_size))
    outputs = hub_module(tf.constant(content_image), tf.constant(style_image))
    stylized_image = tf.reshape(outputs[0], (image_size, image_size, 3))
    file_path = os.path.join(output_path, str(i + 1) + ".jpg")
    img = Image.fromarray(np.uint8(stylized_image * 255.0))
    img.save(file_path)
    elapsed = time.time() - begin
    speed = elapsed / (i + 1)
    total_estimated_time = speed * num_outputs
    if (i + 1) % 100 == 0:
        print("%.2fs / %.2fs %.2f%%" % (elapsed, total_estimated_time, (i + 1) /
 ↪  num_outputs * 100))
```

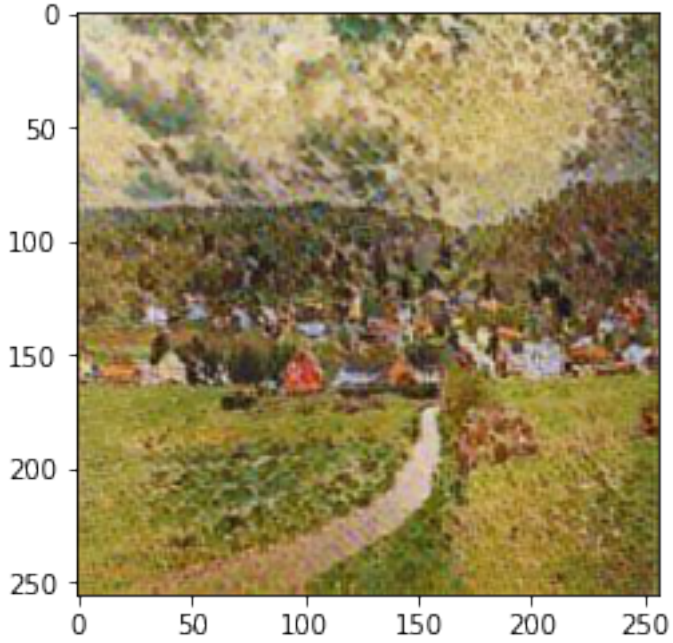
```
66.60s / 4661.74s 1.43%
115.92s / 4057.16s 2.86%
163.92s / 3824.79s 4.29%
208.12s / 3642.12s 5.71%
250.66s / 3509.23s 7.14%
292.46s / 3411.99s 8.57%
331.02s / 3310.15s 10.00%
370.71s / 3243.67s 11.43%
410.25s / 3190.86s 12.86%
448.26s / 3137.83s 14.29%
486.16s / 3093.74s 15.71%
523.45s / 3053.47s 17.14%
559.81s / 3014.37s 18.57%
598.78s / 2993.89s 20.00%
633.93s / 2958.35s 21.43%
670.81s / 2934.78s 22.86%
706.38s / 2908.63s 24.29%
740.54s / 2879.86s 25.71%
773.36s / 2849.24s 27.14%
809.52s / 2833.31s 28.57%
842.79s / 2809.31s 30.00%
878.08s / 2793.88s 31.43%
913.58s / 2780.46s 32.86%
948.50s / 2766.45s 34.29%
984.04s / 2755.31s 35.71%
```

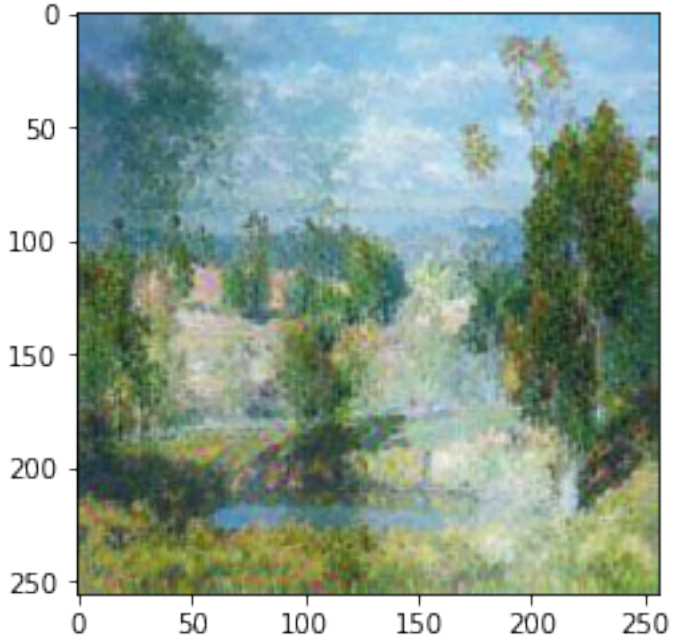
```
1018.40s / 2741.85s 37.14%
1052.09s / 2727.64s 38.57%
1084.12s / 2710.30s 40.00%
1116.73s / 2695.54s 41.43%
1149.02s / 2681.04s 42.86%
1185.95s / 2677.95s 44.29%
1216.23s / 2660.49s 45.71%
1248.94s / 2649.27s 47.14%
1280.17s / 2635.64s 48.57%
1311.92s / 2623.84s 50.00%
1340.61s / 2606.73s 51.43%
1370.55s / 2592.94s 52.86%
1399.95s / 2578.85s 54.29%
1430.89s / 2568.26s 55.71%
1460.33s / 2555.57s 57.14%
1489.51s / 2543.07s 58.57%
1519.00s / 2531.66s 60.00%
1549.44s / 2522.34s 61.43%
1580.13s / 2513.84s 62.86%
1609.00s / 2502.89s 64.29%
1636.81s / 2490.80s 65.71%
1665.15s / 2480.01s 67.14%
1692.96s / 2468.91s 68.57%
1723.49s / 2462.13s 70.00%
1753.42s / 2454.78s 71.43%
1782.22s / 2446.18s 72.86%
1811.31s / 2438.30s 74.29%
1838.37s / 2428.03s 75.71%
1866.59s / 2419.65s 77.14%
1894.43s / 2411.09s 78.57%
1922.74s / 2403.43s 80.00%
1951.42s / 2396.49s 81.43%
1979.93s / 2389.57s 82.86%
2005.67s / 2379.61s 84.29%
2032.73s / 2371.52s 85.71%
2057.20s / 2360.72s 87.14%
2083.24s / 2352.05s 88.57%
2109.13s / 2343.48s 90.00%
2135.85s / 2336.09s 91.43%
2162.67s / 2329.03s 92.86%
2188.52s / 2321.16s 94.29%
2212.53s / 2311.60s 95.71%
2240.67s / 2306.57s 97.14%
2268.43s / 2301.31s 98.57%
2294.86s / 2294.86s 100.00%
```

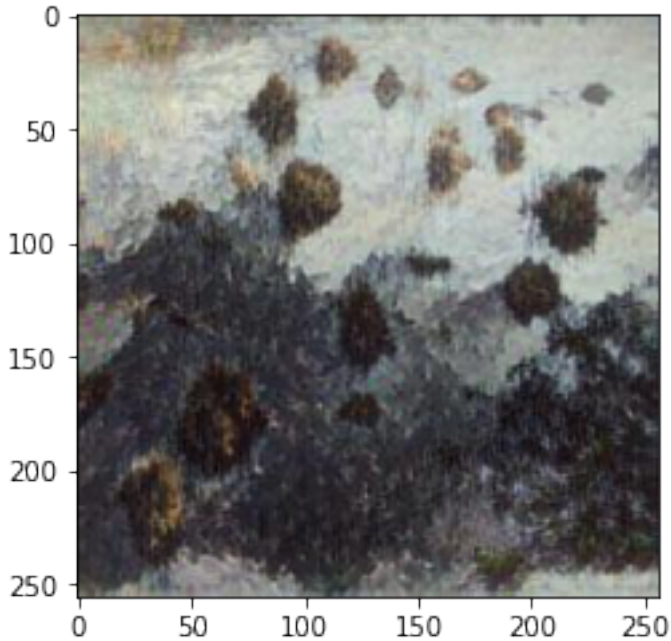
[24]:
```
output_images = np.array([os.path.join(output_path, path) for path in os.
↪listdir(output_path)])
sample(output_images, 5)
```






0
50
100
150
200
250
0
50
100
150
200
250


0
50
100
150
200
250
0
50
100
150
200
250



Reference

https://tfhub.dev/google/magenta/arbitrary-image-stylization-v1-256/2

https://www.kaggle.com/lonnieqin/fast-style-transfer/notebook